GROUP MEETING · PROGRESS UPDATE · NOVEMBER 2026

Two projects in progress

Attribution through compression

Which retrieved sentences did the answer need? One forward pass instead of hundreds.

Re-Cal

Decoding along a reference trajectory: pruned self-speculation, then an edit mode.

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ATTRIBUTION · THE CLASSIC REFERENCE

Which retrieved sentences did the answer need? Shapley credit

$$v(S) = \log p(y \mid x, C_S)$$

How likely the model's own answer y is when only the units in S are visible.

$$\varphi_i = E_\Pi \left[v(P_i \cup \{i\}) - v(P_i) \right]$$

Unit i's gain over the units P_i before it, averaged over every ordering π.

Setting. HotpotQA with distractors: 10 paragraphs, 2 of them gold, about 50 sentence units. Qwen2.5-1.5B-Instruct answers from the full context, and we attribute its own answer.

method	extra passes
Shapley (permutations)	≈ 420
KernelSHAP	256
leave-one-out	n + 1 ≈ 51
ContextCite	32
attention	0
AT2 (learned head weights)	0
ours (learned selector)	**0**

Top four re-run the model on reduced contexts; the rest read the one full-context pass.

Evaluation: the first 200 HotpotQA validation questions · the answer is correct for 111 of them

ATTRIBUTION · WHAT GOES WRONG

Shapley is expensive, and its game rewards confidence in the answer, right or wrong

PROBLEM 1 · COST

≈ 420

forward passes per answer, each on a reduced context.

PROBLEM 2 · THE GAME

v scores how likely the model's **own** answer is. The model is wrong on 89 of 200 questions even with both gold paragraphs present. There the gold evidence argues against the answer, so Shapley scores it as harmful, and look-alike distractors that push toward the wrong answer earn the credit.

49% Shapley **66%** ours

gold sentences kept in the top 10%, wrong answers

> **Example.** "What is the adventure in Tunnels and Trolls?" Gold: Arena of Khazan. The model answers "Crusaders of Khazan", a distractor. Shapley ranks that distractor 1st and 2nd and the gold sentences 6th and 30th. Re-asked with Shapley's top 4: still wrong. With ours: correct.

200 HotpotQA validation questions · the model attends to the gold either way (median attention percentile 94 when wrong)

ATTRIBUTION · OUR METHOD

One learned ordering, read from the pass that wrote the answer, trained on KL

1. **Read.** Attention from the answer to each unit, from the one full-context pass.
2. **Rank.** A small selector (673 parameters linear, or a 0.5M adapter) orders the units; keep the first k.
3. **Train offline** through exact differentiable deletion inside the frozen model, at random budgets k. Zero extra passes at inference.

$$J(\theta) = E_k\,[\, v(\text{top } k \text{ of } \pi\theta)\,]$$

$$= v(\emptyset) + \Sigma_j\, P[k \geq j] \cdot m_{\pi\theta}(j)$$

A budget-weighted marginal vector along **one** ordering. Shapley is the plain average over all n! orderings.

$$v_{KL}(S) = -\,\Sigma_t\, KL(\, p_N(\cdot|y<t) \,\|\, p_S(\cdot|y<t)\,)$$

Keep the model's beliefs, not just one answer.

The identity holds for any game, so the KL-trained selector and the Shapley value of the KL game aim at the **same target**: one ordering, against the average of all.

m_π(j): gain of the j-th unit over the units before it in π · p_N, p_S: next-token beliefs with the full context and with S only

ATTRIBUTION · THE RESULT

Trained on KL, one pass matches the Shapley value of the KL game

5.86 vs 5.83

KL at 10% kept: ours vs KL-Shapley. Paired difference +0.03, 95% CI [−0.32, +0.39].

1 vs 420

forward passes per answer.

0.77 vs 0.66

gold evidence kept in the top 10%: +11 points, 95% CI [+7, +14].

method	passes	KL at 10% ↓	gold kept ↑
Shapley, likelihood game	≈ 420	6.81	0.63
Shapley, KL game	≈ 420	5.83	0.66
AT2	0	6.60	0.73
ours, KL adapter (0.5M)	**0**	**5.86**	**0.77**

200 questions · KL in nats, summed over answer tokens · on the KL game every estimator's KL falls by 0.95–1.19 nats

RE-CAL · THE CONCEPT

Trajectory tracking turns parallel decoding into a supervised problem

1 · PICK A REFERENCE

A slow decoder you trust

LLaDA: one token per pass, most confident first. Nemotron-Labs-Diffusion: its own AR mode, greedy.

→

2 · LABEL FOR FREE

Replay it once

Every masked slot at every canvas the walk visits gets a label: does the token it would commit now differ from the one the reference ends with?

→

3 · COMMIT ON A BUDGET

Stay on its path

Predict $\hat{p}_i \approx P[\text{slot } i \text{ disagrees}]$ from the one forward pass, and commit slots while $\Sigma\, \hat{p}_i \le \varphi$.

PIG and Re-Cal track the sequential decode on LLaDA. On Nemotron the reference is the AR mode, and **self-speculation** is trajectory tracking with an exact verifier. The rest of this part makes that loop faster.

RE-CAL ON NEMOTRON-LABS-DIFFUSION-3B · SELF-SPECULATION

Quadratic self-speculation, pruned to a single candidate block

LINEAR SS · RELEASE LOOP

Draft a block in one forward, verify it with the AR in a second: j + 1 tokens per two forwards.

3.22 / **4.12** TPF · 2 × 32 positions

QUADRATIC SS

Verify, and in the same forward draft a block after every possible accept length c = 0…31: j + 1 tokens per forward.

4.58 / **5.90** TPF · 1,047 positions

PRUNED · C = 1

A PIG-style adapter predicts the accept length, $S(c) = P[j \geq c]$; draft only one block, after c_1.

3.26 / **4.25** TPF · 57 positions

If $c_1 \leq j$, the commit stops at c_1

The rule committed $s_1 \ldots s_c_1$ and the AR's next token, then continued from that block: verified tokens past it were thrown away.

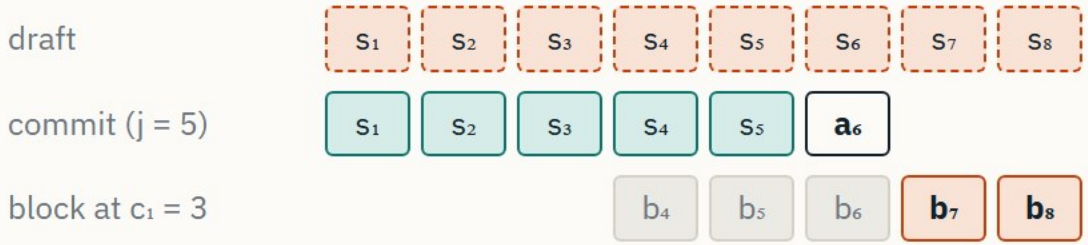
If $c_1 > j$, a miss costs a forward

No block sits at j, so the next forward only drafts. One candidate misses on 26–29% of steps, and the gain over linear SS is gone.

TPF on HumanEval / GSM8K-500, every forward counted · strict verification: the output is the AR greedy walk

RE-CAL ON NEMOTRON · FIXING C = 1

Keep every verified token, and take the next draft from the anchor block



Block 3 was drafted in the same forward, from $s_1 \ldots s_3$. Its guesses past a_6 are appended as the next draft: 3 tokens stale instead of 6.

Always commit j + 1. When $c_1 \le j$, keep all of $s_1 \ldots s_j$ and a_{j+1}, not just the tokens up to c_1.

A miss costs no forward. A block at c = 0 is always drafted too (Packed-SS); if $j < c_1$ it takes over.

Choose the anchor. c_1 = argmax c · S(c): the expected freshness gained.

Running now: the truncation fix alone, then the merged rule.

RE-CAL · THE EDIT MODE (EDLM)

One forward verifies the prefix and edits the tail

CONTEXT

Prompt and committed tokens, cached by the frozen base

AR ROWS · GATE OFF

`x s₁ s₂ … s_m`

Causal, exactly the shipped verifier

Out: AR tokens $a_1 \ldots a_{m+1}$, so the exact accept length j. Commit $s_1 \ldots s_j$ and a_{j+1}.

EDITOR ROWS · GATE ON

`s₁ s₂ … s_m`

A copy of the draft, fully visible, same position ids

Out: per slot, p_ℓ over the vocabulary: keep $\kappa_\ell = p_\ell(s_\ell)$, edit = argmax p_ℓ. The edited tail is the next draft.

Inputs

The cached context, the drafted block shown in full, and the AR stream over the same draft.

Model

A per-token-gated LoRA on every layer (r = 64, about 99M parameters); base and LM head frozen.

Gate off

Every projection returns the base output: AR, diffusion and self-speculation are bit-identical.

RE-CAL · TRAINING THE EDIT MODE, NOW

Train it to predict the AR's token at every drafted slot

LOSS

$$L(\theta) = E[\ -\ \Sigma_\ell\ w_\ell\ \log\ p_\theta(x_\ell \mid z, c)\ ]$$

$$w_\ell = \exp(-d_\ell / \gamma),\ \gamma \in \{8, 15, \infty\}$$

z: the drafted block. c: the cached context. x: the AR greedy tokens at the block's positions, free from the lossless loop. dℓ: distance from the commit point. Drafted slots only; keep = predict the shown token.

Optimum: κℓ = 1 − P[slot disagrees | features], Re-Cal's calibrated reader, plus an edit.

THE LOOP

1. The diffusion mode drafts a block.
2. One forward: AR rows verify, editor rows read the draft.
3. Commit s₁…s_j and a_{j+1}: lossless.
4. Accept any subset of the tail; replace or re-mask the rest.
5. The repaired tail is the next draft; go to 2.

Data. States from the lossless anchored C = 1 loop, plus loop-free blocks of 16, 32 or 64 slots, on 2,000 Nemotron SFT prompts.
Next: next-draft acceptance offline: edited tail vs unedited tail vs the loop's redraft.

BACKUP · THEORY

What the sufficiency ordering keeps, gives up and gains against Shapley

KEEPS

Efficiency

Null player

Affine invariance

GIVES UP, BY DESIGN

Symmetry

Additivity

Marginalism

GAINS

Robustness to duplicated units

Sufficiency optimality

(1 – 1/e) for submodular games

The two are in one-to-one correspondence, and they coincide exactly when the game is additive. Duplicating a sentence splits its Shapley credit; the ordering keeps one copy and moves on.

BACKUP · THE FULL COMPARISON

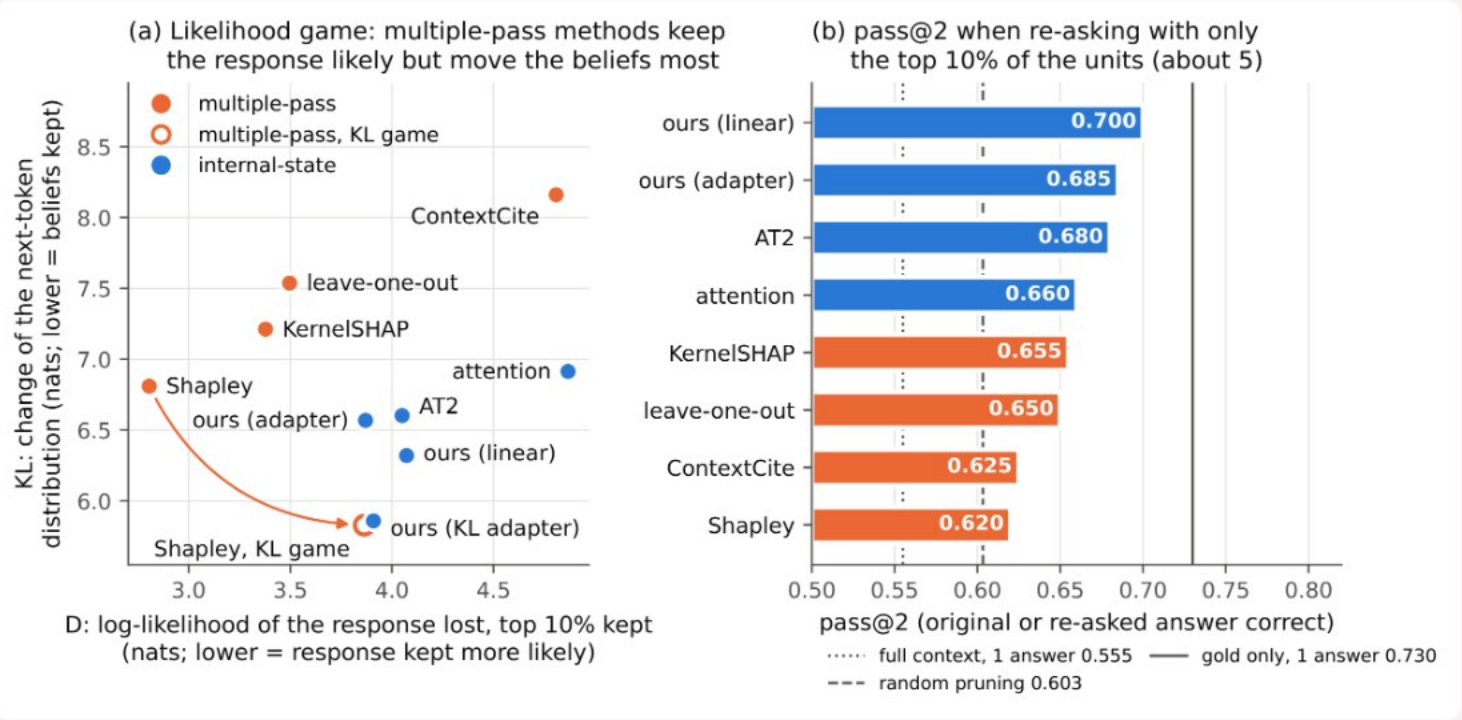
Every method on both games, at 5% and 10% kept

method	passes	D 5%	D 10%	KL 5%	KL 10%	gold all	right	wrong
Shapley · likelihood	≈420	5.67	2.81	9.20	6.81	0.63	0.75	0.49
KernelSHAP · likelihood	256	5.76	3.38	9.28	7.21	0.59	0.68	0.48
leave-one-out · likelihood	n+1	6.71	3.50	10.22	7.54	0.59	0.67	0.49
ContextCite · likelihood	32	7.33	4.81	10.60	8.16	0.50	0.58	0.41
Shapley · KL	≈420	6.24	3.86	8.57	5.83	0.66	0.75	0.55
KernelSHAP · KL	256	6.41	3.98	8.74	6.02	0.64	0.72	0.54
leave-one-out · KL	n+1	7.26	4.84	9.35	6.59	0.71	0.78	0.62
attention	0	7.51	4.87	9.81	6.91	0.69	0.76	0.62
AT2	0	6.16	4.05	9.07	6.60	0.73	0.84	0.61
ours, linear	0	6.61	4.07	8.90	6.32	0.76	0.84	0.66
ours, adapter 2.1M	0	6.77	3.87	9.53	6.57	0.74	0.84	0.62
ours, KL adapter 0.5M	**0**	6.40	3.91	8.87	**5.86**	**0.77**	**0.84**	**0.67**

D and KL in nats, lower is better · gold = share of gold sentences in the top 10%, split by whether the answer was right (111) or wrong (89)

BACKUP · WHY THE LIKELIHOOD SCORE MISLEADS

Keeping the answer likely is not keeping the model's beliefs



$$KL_t = p_N(y_t) \cdot \Delta(y_t) + \Sigma_{w \neq y_t} p_N(w) \cdot \Delta(w)$$

$$\Delta(w) = \log p_N(w) - \log p_S(w)$$

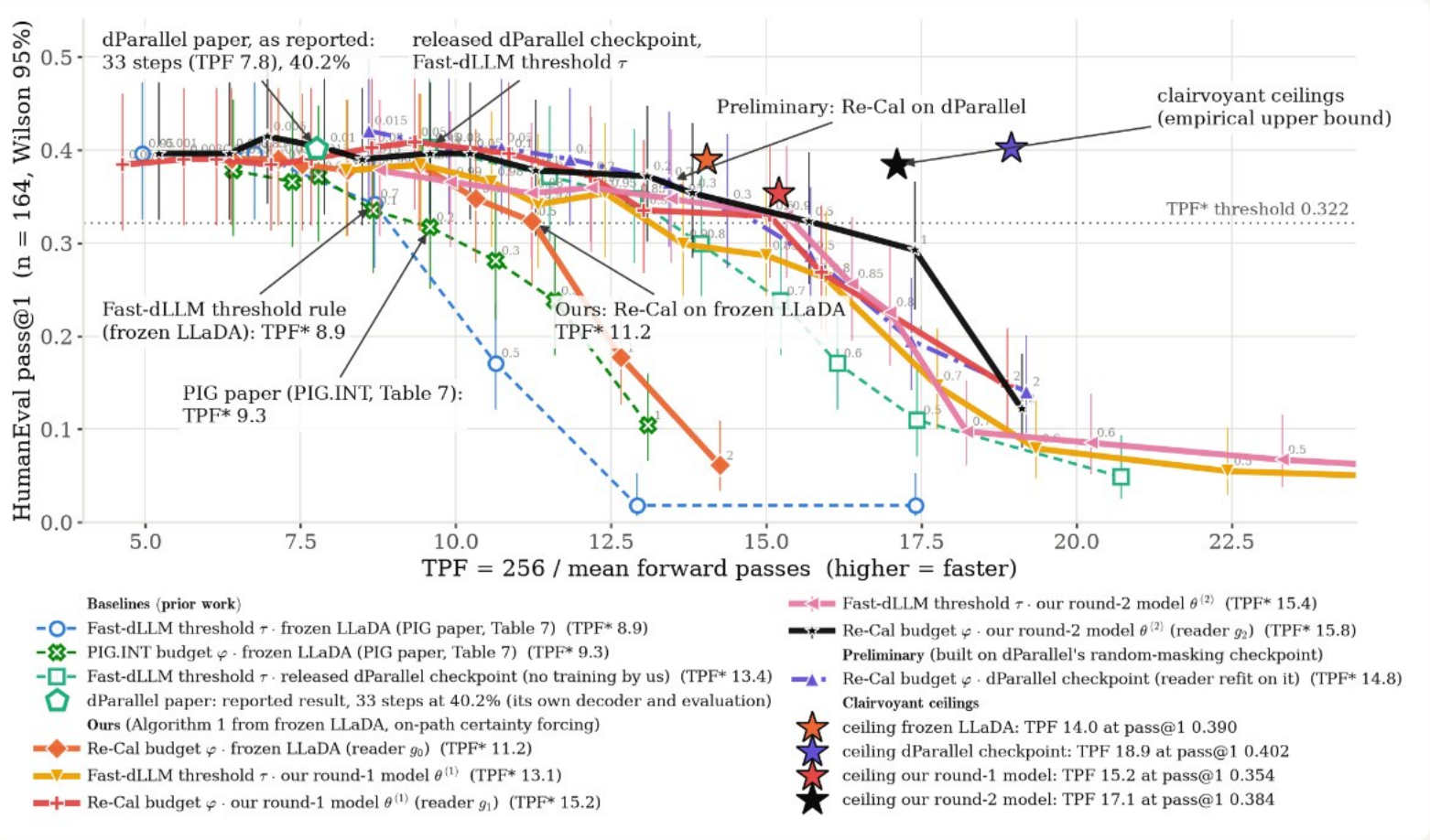
A kept set that makes the model **more** confident in its answer lowers the first term and D, even below zero, but drains the alternatives it was weighing and raises the second.

Querying the model finds such sets. Reading its attention cannot.

200 questions, 10% kept · right panel: pass@2 = the original or the re-asked answer is correct; gold paragraphs alone 0.730

BACKUP · RE-CAL ON LLADA-8B-INSTRUCT

PIG's reader without the pairwise estimator, then two rounds



$p_i \approx P[x_i \neq \text{final}]$

PIG.INT, frozen

9.3

Re-Cal, frozen

11.2

Re-Cal, round 2

15.8

HumanEval, TPF* (fastest setting with pass@1 ≥ 0.322) · stars: clairvoyant ceilings, 14.0 → 17.1 after two rounds

BACKUP · PRUNING WITH MORE CANDIDATES

Five candidates keep most of quadratic self-speculation at a fifth of its width

decoder	positions	miss %	HumanEval	TPF	GSM8K-500	TPF
linear SS (release)	2 × 32	–	0.793	3.22	0.892	4.12
quadratic SS, all 32	1,047	0	0.780	4.58	0.894	5.90
pruned, C = 1	57	29 / 26	0.780	3.26	0.898	4.25
pruned, C = 3	121	7 / 6	0.805	3.98	0.890	5.16
pruned, C = 5	**187**	2.5 / 1.9	0.799	**4.27**	0.890	**5.53**

The rule

Each step keeps the C blocks that maximise expected tokens over expected forwards, under the adapter's accept-length law.

What it buys

C = 5: +32% / +34% TPF, lossless, 76–85% of the quadratic gain. It beats tuned offsets by 4–6%.

Nemotron-Labs-Diffusion-3B · strict verification · paired 95% accuracy intervals vs linear SS contain 0 for every pruned row